Mission Report #213 for the 2026 LASC

Team Serra Rocketry - Mission Helike

Abstract

Helike #213 is a 1P PocketQube satellite developed by Serra Rocketry (IPRJ-UERJ) for the 7th Latin American Space Challenge (LASC 2026). Its central subsystem is the SRAB (Bioinspired Autorotative Recovery System) — a passive, parachute-free recovery mechanism inspired by samara seed autorotation that eliminates the electromechanical sequencers and pyrotechnics responsible for conventional parachute failures. Three iterative simulation-and-test campaigns validated the aerodynamic model, and the full descent from the Dédalo launch vehicle apogee (1520.5 m AGL) was simulated with RocketPy under real GFS conditions, predicting a terminal velocity of 13.33 m/s with a 1.5 safety factor over the LASC 20 m/s limit. A LoRa telemetry link with RSSI trilateration supports post-landing recovery.

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Nomenclature

BMS	= Battery Management System
BEM	= Blade Element Method
ConOps	= Concept of Operations
EPS	= Electrical Power System
FMECA	= Failure Mode, Effects, and Criticality Analysis
GPS	= Global Positioning System
IMU	= Inertial Measurement Unit
LASC	= Latin American Space Challenge
LEV	= Leading-Edge Vortex
LoRa	= Long Range wireless communication protocol
MCU	= Microcontroller Unit
PQB	= PocketQube
RBF	= Remove Before Flight
RSSI	= Received Signal Strength Indicator
SCSM	= Satellite Challenge Standards Manual
RTOS	= Real-Time Operating System
SRAB	= Sistema de Recuperação Autorrotativo Bioinspirado (Bioinspired Autorotative Recovery System)
θ	= conicity angle
$\dot{\phi}$	= rotor spin rate
C_L	= lift coefficient
C_D	= drag coefficient
C_{D0}	= parasitic (flat-plate) drag coefficient
α	= local angle of attack
ρ	= air density
v_0	= vertical descent velocity
r	= radial position along the wing
U	= local resultant airspeed
V_F	= impact (terminal) velocity
SF	= safety factor
R	= risk score ($R = P \times S$)
P	= failure probability
S	= failure severity

1. Introduction

A. Mission Overview and Motivation

Team Serra Rocketry, based at the Instituto Politécnico (IPRJ-UERJ), is developing the Helike #213 1P PocketQube for the Satellite Challenge category of the 7th Latin American Space Challenge (LASC 2026). The mission carries the weight of two previous missions in the competition and the determination not to repeat the mistakes that cost the team everything.

For the team, LASC is not an end in itself but the means: the flight opportunity it provides is the primary in-flight test environment for the SRAB recovery technology. The mission is therefore framed as a technology-development flight test — the satellite exists to carry the SRAB through a real descent and return instrumented evidence of its behavior, so that a recovery concept born in simulation can graduate to flight-proven technology. Success is measured by the quality of the validation data recovered, not by a competitive result.

The LASC 2025 Lesson

At LASC 2025, Serra Rocketry flew two missions: the SR Couto rocket (#100) and the SR Coutinho satellite (#261). Both were built with self-funded resources, unique components, and no redundancy. One day before launch, the onboard electronics began to fail — a burned BMP and an unstable LoRa link. With no redundancy and no backup; the flight was spectacular but communication was lost, the parachute did not deploy, and the flight ended ballistic. Both rocket and payload were lost.

The lesson crystallized: *optimization without redundancy is a trap*. The Helike is the materialization of this lesson.

B. Form Factor Selection: Why a 1P PocketQube?

Among the form factors available to university satellite teams, the Helike mission deliberately selects the PocketQube 1P standard (50×50×50 mm, ∼200 g) over the two common alternatives: CanSats and CubeSats. A CanSat offers no standardized deployment envelope and no heritage in the deployer ecosystem, and its size and mass do not exercise the recovery problem that defines this mission. A 1U CubeSat (10×10×10 cm, ∼1 kg), at the opposite extreme, would consume a disproportionate share of the Dédalo launch vehicle payload allocation, leaving little room for the recovery validation that is the core of the mission. The 1P PocketQube sits at the right point of the trade-off: the smallest standardized satellite form factor, with a ∼200 g mass that fits comfortably within the 800 g payload class of the Dédalo rocket.

The choice was enabled by a direct donation from Alba Orbital, received at LASC 2025: a flight-grade 1P frame consisting of an aluminum structure with a fiber-reinforced polymer (FRP) backplate. The aluminum structure provides the rigidity required to survive launch vibration and the ejection shock at the lowest mass compatible with the 50 mm envelope, while the non-conductive FRP backplate provides an isolated mounting surface for the electronics stack, simplifying the electrical design. By removing the need to develop and qualify a mechanical structure in-house, the donation allowed the team to concentrate development effort on the SRAB — the subsystem that defines the mission.

C. The Problem with Conventional Parachutes

The LASC 2025 experience only confirmed that the parachute is a critical failure point: a conventional parachute depends on a chain of events: a sequencer decides when to open, an actuator (servo or pyrotechnics) fires, and enough energy must be available at exactly the right moment. Break any link in that chain and the parachute never opens, which means total loss.

D. The SRAB Solution

The SRAB (Bioinspired Autorotative Recovery System) was born directly from the pain of losing everything: *"what if recovery simply... worked? No servo, no pyrotechnics, no software decision — just physics."*

The system is fully passive: no actuators, no deployment electronics, no mechanism at all. Recovery happens by aerodynamic principle — the physics does the work, not a decision made in software. Because the descent is governed by rigid-body dynamics, the behavior is predictable and modelable. And the wings stay stowed during flight, deploying passively at ejection, so the whole mechanism fits inside the deployer envelope.

E. Bioinspired Innovation

The SRAB is based on a mechanism that nature has refined over millions of years: the **autorotation of samara seeds** (*Acer rubrum*, maple seeds) [1,2]. These seeds evolved an asymmetric geometry where the center of mass sits at one end and the aerodynamic lift at the other, inducing stable conical rotation during fall. The Leading-Edge Vortex (LEV) generated by the rotation doubles the effective lift [1], resulting in very low descent velocities for the mass of the object.

Adapting this biological solution to aerospace engineering for PocketQubes is a step toward simpler, more reliable, and more elegant missions.

F. Design Philosophy: The Three-Filter Review

The project is reviewed continuously through a **three-filter philosophy** that links regulatory compliance, failure analysis, and schedule discipline directly to the code and hardware being built:

Filter 1 — Compliance (SCSM LASC 2026)

Every design decision is checked against the Satellite Challenge Standards Manual (SCSM, Edition 7, Revision 1) [5]. Compliance is organized in four themes. *Electrical isolation* comes first: two independent mechanical kill switches (SW1/SW2, NC contacts wired in parallel, Z axis) keep the satellite powered off during integration inside the deployer, meeting PQB 7.1.9–7.1.11, and the RBF pin (SW3), in series with the main power rail, cuts all energy when inserted and is removed only after integration (ECS 9.1.5–9.1.7). The satellite has no fixed electrical connection to the launch vehicle (REC 10.1.1); it powers on at ejection and starts telemetry immediately. *Energy* follows: the power budget (Section 3) predicts about 8 h of autonomy against the ECS 9.1.3 requirement of at least 4 h — a margin of two times — with a bench test at 20% margin as the remaining evidence. *Recovery* is the heart of the mission: because the SRAB is a non-parachute recovery system, REC 10.2.1 requires that LASC be notified; the notification was sent on First Progress Update. Finally, *spectrum*: LoRa runs at 915 MHz, the legal Americas band in Brazil (ECS 9.2.4).

Filter 2 — Murphy (What Can Go Wrong)

Each Single Point of Failure (SPoF) is identified and mitigated in *both* hardware and firmware. The highest risk is SRAB deployment failure, and the design attacks it from two sides. Physically, the system is passive with no hinges to jam, and three drop-test campaigns validated deployment (Section 4). In firmware, the strategy is the opposite of complexity: the flight software runs a single synchronous polling loop that samples, validates and transmits at 20 Hz and a hardware watchdog (TWDT, 5 s) resets the system if the loop ever stalls. Energy failure is covered by a robust BMS with overcharge, overdischarge, short-circuit and overcurrent protection. Data are protected twice over: if LoRa is lost, the SD card preserves every byte of flight data and if the SD card itself fails, the firmware falls back to LittleFS on the internal flash. Because the LoRa link is transmit-only, there is no uplink or in-flight command path; recovery never depends on a software decision.

Filter 3 — Timeline (Deadlines)

The schedule is enforced through milestone-driven development with the LASC progress updates as hard gates and the competition in September 2026. Every deliverable is mapped to an owner and an evidence artifact (video, log, bench test) so that the filter reviews are data-driven rather than opinion-driven.

G. Mission Objectives

Primary Objective — SRAB Validation

Demonstrate that the SRAB is a viable recovery mechanism for 1P PocketQubes, with a predictable terminal velocity and no single critical failure points .

Secondary Objective — Post-Landing Location

Use LoRa RSSI trilateration (3 ground beacons) to locate the satellite on the ground after landing, complementing SRAB recovery.

Support Objective — Flight Telemetry

Collect flight data (pressure, temperature, acceleration, rotation, GPS position) and transmit via LoRa, supporting the mission by documenting the descent and validating the SRAB model.

H. Project Specifications

The principal design specifications of the Helike #213 are summarized in Table 1.

2. System Architecture and Concept of Operations

A. System Architecture

Helike #213 is a PocketQube 1P (50×50×50 mm, ∼200 g) whose central subsystem is the SRAB — the bioinspired autorotative recovery system. The satellite is organized in five functional blocks:

Mechanical Structure

The flight structure is the 1P frame donated by Alba Orbital, machined in aluminum with a fiber-reinforced polymer (FRP) backplate. The aluminum structure carries the launch vibration and ejection shock loads with minimum mass,

Parameter	Value
Form factor	PocketQube 1P (50×50×50 mm)
Recovery	SRAB — 2 wings, passive, non-parachute
Terminal velocity (predicted)	13.33 m/s ($V_F = 20$ m/s target, SF 1.5)
Microcontroller	ESP32-C3 Super Mini (single-core RISC-V)
Communication	LoRa RFM95W, 915 MHz, SF7, 125 kHz bandwidth, 20 dBm, sync 0xF3, CRC enabled
Telemetry rate	20 Hz
Navigation	GPS NEO-8M
Sensors	BME280 (pressure/temp/humidity) + ICM-20602 (6-axis IMU)
Storage	SD card + LittleFS fallback
Safety	2 kill switches + RBF pin
Battery	Li-Ion 1300 mAh
Autonomy	≈8 h (margin ≈ 2×)

Table 1 Project specifications.

and the FRP backplate provides an isolated, non-conductive mounting plane for the electronics stack, eliminating the need for additional standoffs or insulating hardware. The mechanical architecture follows the PocketQube 1P standard envelope, guaranteeing compatibility with the deployer and with the Dédalo launch vehicle integration.

B. Subsystem Descriptions

Recovery (SRAB)

The SRAB is a passive recovery system composed of 2 autorotative wings, stowed during flight and deployed by ejection. They convert the potential energy of the fall into rotational kinetic energy, slowing the satellite to a predicted terminal velocity of 13.33 m/s — the $V_F = 20/1.5$ m/s design target, applying a 1.5 safety factor to the LASC 20 m/s limit. There are no actuators: no servo, no pyrotechnics, no deployment electronics. Recovery does not depend on software at all, so it works even with total loss of the avionics. The two-wing configuration keeps the mechanism simple and light; losing a single wing is a mission-critical degradation, so the design relies on the robustness of the geometry and on the drop-test evidence accumulated across the tests iterations. The wing geometry and hub assembly used in the computational simulation and drop tests are shown in Fig. 1.

Electrical Power System

Energy comes from a 1300 mAh Li-Ion battery protected by a BMS that covers overcharge, overdischarge, short-circuit and overcurrent. With an average draw of about 145 mA, the predicted autonomy exceeds 8 h, a margin of roughly two times the SCSM requirement of 4 h. Safety is handled by two kill switches (SW1/SW2, NC contacts wired in parallel, Z axis) plus the RBF pin (SW3), which cuts all energy when inserted. The satellite only powers on at ejection.

Avionics and Telemetry

At the core sits an ESP32-C3 Super Mini, a single-core RISC-V microcontroller running a single synchronous 20 Hz polling loop. The ESP32-C3 was selected over the dual-core ESP32-S3 used in the team's rocket flight computers: the flight loop reads two sensors, assembles a telemetry packet and transmits it at 20 Hz, a workload that keeps a single 160 MHz RISC-V core below a few percent duty cycle. A second core would add idle current and cost without contributing capability, and the bare polling loop — no RTOS, no preemption — eliminates entire classes of concurrency faults from the recovery path. The integrated Wi-Fi and Bluetooth radios are kept disabled to minimize energy draw, leaving LoRa as the only active radio.

A BME280 measures pressure, temperature and humidity, and an ICM-20602 six-axis IMU adds acceleration and angular rate; together they feed the telemetry packet and the SRAB rotation analysis. The BME280 was chosen for the mission's energy profile: in measurement mode it draws 3.6 μA at a 1 Hz refresh rate, negligible against the link budget, and a single I2C transaction returns all three environmental quantities. The ICM-20602 was chosen for its low-power MEMS architecture and its sampling rate: with the SRAB autorotating at roughly 8 Hz at equilibrium (477 rpm), the 20 Hz polling loop sits above the Nyquist frequency of the spin, so the IMU resolves the rotation without aliasing.

Telemetry goes down through the LoRa RFM95W at 915 MHz, transmit-only, while a GPS NEO-8M records the position. The link operates at spreading factor SF7 with 125 kHz bandwidth, the lowest-airtime combination in the LoRa family: for a fixed packet size, SF7 minimizes on-air time, which directly reduces the energy per transmission — the radio is the highest-draw component at 120 mA during bursts — and the channel occupancy, lowering collision risk with other teams on the 915 MHz band. Transmit-only operation removes the receiver entirely: there is no uplink, no in-flight command path and no remote attack surface, and recovery never depends on a software decision, as discussed in Section 1 (Filter 2).

Flight data are written to an SD card with automatic LittleFS fallback on the internal flash, and three ground beacons measure RSSI for trilateration once the satellite is on the ground.

C. Concept of Operations

From a technology-development standpoint, the ConOps doubles as the flight-test plan for the SRAB: each arc exercises one stage of the recovery system — powered-off safety while stowed (SAFE), ejection and power-on (DEPLOY), passive deployment and autorotation (DESCENT), and ground location (POST-LANDING) — and the descent telemetry is the primary evidence for validating the aerodynamic model against real flight data.

The satellite's behavior follows the physical flight profile in four arcs. The first arc is *inert*: the satellite starts in SAFE, with the kill switches and the RBF pin inserted, and stays completely powered off until ejection. The second arc begins at *ejection*: the satellite powers on, initializes the sensors and attempts a first GPS fix. The third arc is the *descent*: at apogee the satellite leaves the rocket, the two SRAB wings deploy passively — no actuator, no command required — and the descent is governed entirely by autorotation while the telemetry packet streams rotation rate, descent velocity. The fourth arc is *post-landing*: impact detection, the current firmware simply keeps transmitting periodic packets with RSSI and GPS position until the battery is exhausted or the recovery team arrives.

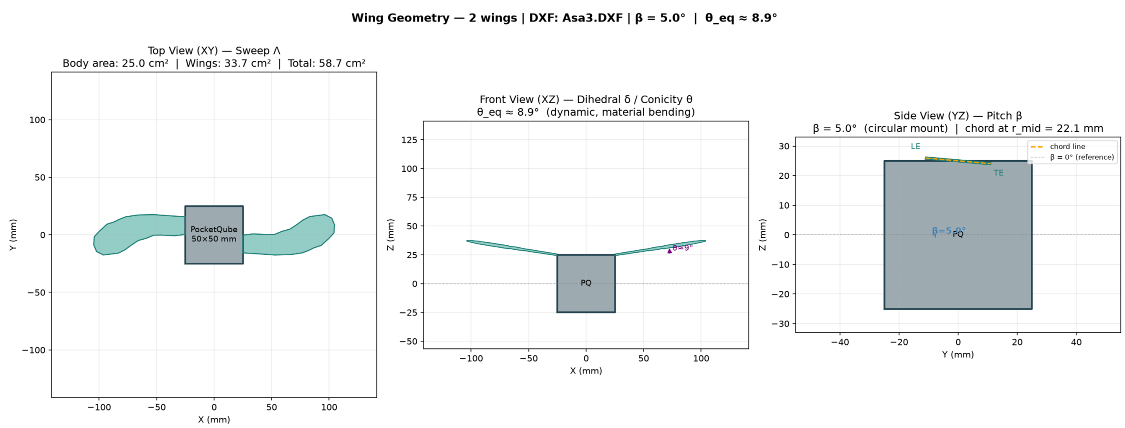

Fig. 1 SRAB wing geometry views: three orthogonal projections of the wing + hub assembly used in the computational simulation and drop tests.

3. Weights, Measures, and Performance Data

A. Physical Characteristics

The principal physical characteristics of the satellite are listed in Table 2.

Parameter	Value
Form factor	PocketQube 1P (50×50×50 mm)
Mass	∼200 g
Stowed frontal area	∼25 cm^2
Deployed frontal area (SRAB, 2-wing)	∼56 cm^2

Table 2 SRAB physical parameters.

B. Recovery Performance

The SRAB recovery performance was evaluated through three iterative simulation-and-test campaigns and computational simulation (Section 4). The consolidated recovery performance data are summarized in Table 3:

Parameter	Value
Terminal velocity (final, Asa3 2-wing)	13.33 m/s (simulated from real apogee)
Impact velocity (Test 2, 20 m drop, Asa2)	8.04 m/s
Rotation rate (Test 2, Asa2)	306.9 rpm
Cone angle (Test 2, Asa2)	24.4°
LASC descent criterion	$V_F = 13.33$ m/s (20 m/s / SF 1.5)
Impact energy dissipated by autorotation	99.15%
REC 10.2.1 notification (non-parachute)	Sent (1st + 4th Progress Updates, email)

Table 3 SRAB recovery performance data.

C. Power and Autonomy

The power budget and predicted autonomy are presented in Table 4:

Parameter	Value
Battery	Li-Ion 1300 mAh
Average current draw	$\sim$145 mA
Predicted autonomy	$>$8 h
SCSM requirement	$\geq$4 h
Margin	$\approx 2\times$

Table 4 Power budget and predicted autonomy.

D. Flight Sensors

The flight sensors and their measurements are listed in Table 5:

Sensor	Type	Measurement
ICM-20602	6-axis IMU	acceleration + angular rate
BME280	Environmental	pressure, temperature, humidity
NEO-8M	GPS	position, velocity, time
RFM95W	LoRa radio	915 MHz telemetry link

Table 5 Flight sensors and measurements.

4. Computational Simulation

A. Theoretical Background

Samara Autorotation

The SRAB is inspired by the autorotation of samara seeds (*Acer rubrum*). The seed geometry places the center of mass at one end and the aerodynamic center at the other, generating a torque imbalance that produces stable conical rotation during fall. The resulting motion is a steady autorotation state where the descent rate is substantially lower than the ballistic terminal velocity of the same mass [2].

Leading-Edge Vortex (LEV)

At high incidence angles and rotation rates, a stable leading-edge vortex forms on the wing upper surface, increasing the effective lift coefficient beyond the thin-airfoil limit. This LEV effect is the key to the samara's efficiency and is modeled through an augmented lift coefficient in the blade element formulation.

Reduced-Order Rigid-Body Model

The descent is modeled with a 4th-order reduced Newton-Euler formulation [3], with state vector (Eq. 1):

$$\mathbf{x} = [\theta,\ \dot{\theta},\ \dot{\phi},\ v_0]^T \tag{1}$$

where θ is the conicity angle, $\dot{\phi}$ the rotor spin rate and v_0 the vertical descent velocity. The aerodynamic loads are computed with the Blade Element Method (BEM) (Eq. 2):

$$dF = \frac{1}{2}\rho\, w(r)\, \|U\|^2 \,[C_L(\alpha),\, C_D(\alpha)]\, dr \tag{2}$$

with the thin-airfoil lift curve and the induced-drag model (Eq. 3):

$$C_L = 2\pi \sin\alpha, \qquad C_D = C_L \sin\alpha + C_{D0} \tag{3}$$

where $C_{D0} = 1.0$ is the flat-plate drag coefficient and the LEV contribution is embedded through the $f_{factor} = 0.3$ correction in the lift build-up [4].

B. Simulation Methodology

The validation pipeline is an iterative loop: model, simulate, prototype, drop, compare, and iterate on the geometry. We start from a 4th-order rigid-body dynamics model with a Blade Element Method aerodynamic model, implemented in RocketPy [6] with SRAB extension modules. The descent is then simulated from the real apogee of the launch vehicle, the Dédalo rocket, in the same GFS atmosphere and with the same payload mass as the actual flight. PETG 3D-printed wings (Asa1, Asa2 and Asa3 geometries) are dropped from controlled heights with instrumentation, and the measured terminal velocity, spin rate and cone angle are compared against the simulation. The wing geometry is updated between test cycles based on that comparison, which is how the design improved from one campaign to the next.

Launch Vehicle Model — Dédalo

The Dédalo rocket (used as the launch vehicle for Helike) is modeled in RocketPy, with the parameters summarized in Table 6:

Parameter	**Value**
Total mass (with payload)	5.100 kg
Empty mass (Stage 2)	4.900 kg
Payload mass (Helike #213)	0.200 kg
Diameter	106 mm
Motor	SolidMotor, 926 N peak, 2.82 s burn
Motor data points	95
Atmosphere	GFS (real, launch-day forecast)
Launch site	lat −21.94305, lon −48.95409, elev 478 m

Table 6 Dédalo #11 simulation parameters.

C. Results

Dédalo Ascent (with payload)

The simulated ascent of the Dédalo launch vehicle carrying the Helike payload is summarized in Table 7:

Parameter	Value
Apogee	1520 m AGL (1996 m ASL)
Time to apogee	17.83 s
Maximum velocity	205.3 m/s (0.59 Mach)
Maximum acceleration	106.7 m/s^2 (10.9 g)
Rail departure	0.47 s at 18.9 m/s

Table 7 Dédalo #11 flight simulation results.

SRAB Descent from Real Apogee

The SRAB simulation is initialized at the real apogee of Stage 1 (1520.5 m AGL, $v_z \approx 0$), with the optimized wing radius for the target impact velocity $V_F^{target} = V_{F,max}/FS = 20/1.5 = 13.33$ m/s. The descent results are summarized in Table 8, and the complete descent profile is presented in the LRR certification dashboard (Fig. 2) and the satellite-view trajectory map (Fig. 3):

Parameter	Value
Wing geometry	Asa3 (2 wings), optimized radius 79.7 mm
Release altitude	1520.5 m AGL
Impact velocity	13.33 m/s
Descent time	111.4 s
Equilibrium cone angle θ_{eq}	8.6°
Impact spin rate	477 RPM
Total drift	184.3 m
Impact position (relative)	x = −67.5 m, y = +171.5 m
LASC descent criterion	Target $V_F = V_{F,max}/FS = 20/1.5 = 13.33$ m/s; simulated impact 13.33 m/s (PASS, FS 1.5)

Table 8 SRAB descent simulation results from real apogee.

SRAB vs. Parachute Comparison

A flat circular parachute (∅200 mm, $C_D = 1.5$) was simulated in the same GFS atmosphere, same release conditions and same payload mass, as a benchmark. The comparison is summarized in Table 9:

Parameter	SRAB	Parachute	Ratio
Impact velocity (m/s)	13.33	8.48	1.57×
Descent time (s)	111.4	174.6	0.64×
Total drift (m)	131.2	204.6	0.64×
Impact energy (J)	17.8	7.2	2.47×
Terminal velocity (SL) (m/s)	13.33	8.24	1.62×

Table 9 SRAB vs. parachute comparison.

The SRAB achieves a 36% shorter descent time and 36% lower drift than the parachute, at the cost of a higher impact velocity and energy — both still well below the 20 m/s LASC limit. The shorter descent reduces wind exposure (drift), which is the dominant factor for recovery accuracy on miniaturized satellites.

Monte Carlo Analysis

A Monte Carlo analysis with 100 iterations of the Asa3 2-wing configuration, released from the real simulated apogee, assessed the descent under atmospheric dispersion: the impact velocity stayed at 13.33 ± 0.31 m/s (P5 12.78, P95 13.87), the descent time at 110.97 ± 2.53 s, the spin rate at 475.9 ± 12.8 rpm and the equilibrium cone angle at $8.63 \pm 0.67°$ — every iteration below the 20 m/s LASC limit, confirming the 1.5 safety factor.

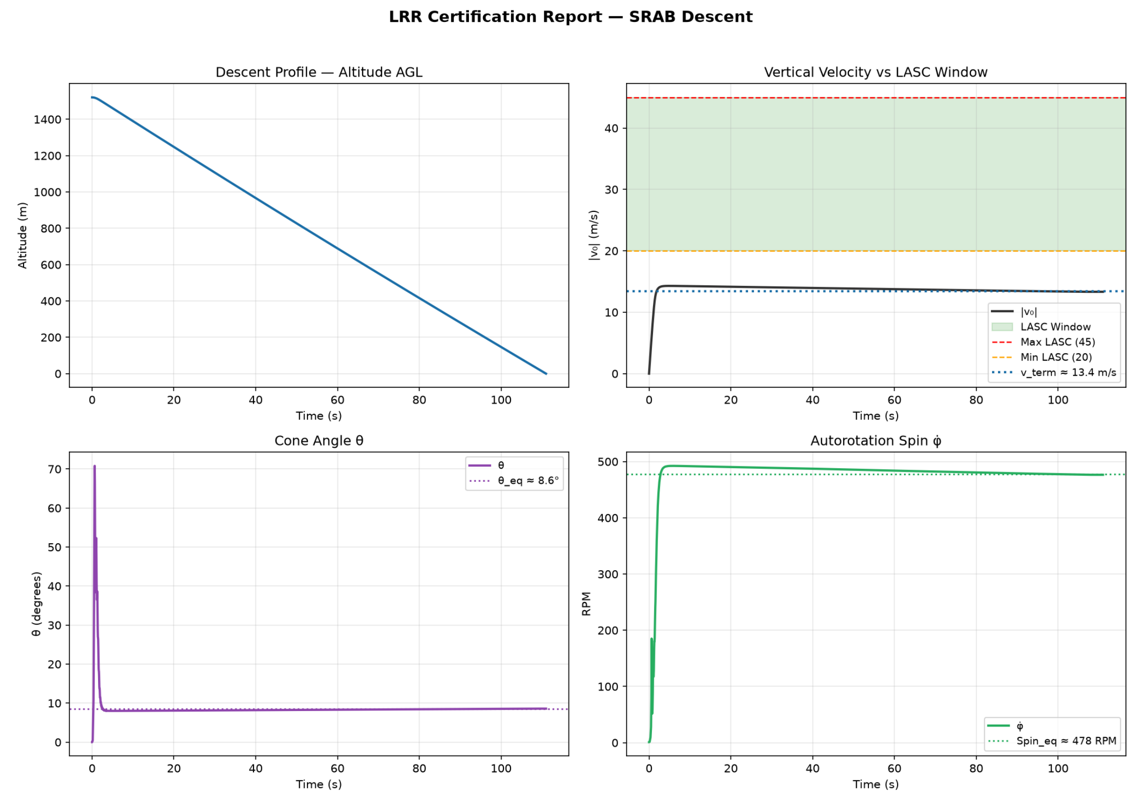

Fig. 2 LRR certification dashboard for the SRAB descent from the real apogee: altitude profile, vertical velocity, cone angle θ, and autorotation spin rate.

Fig. 3 Satellite view of the simulated SRAB descent trajectory over the Dédalo launch site: release point (green marker) at apogee and predicted impact point (red marker, 100 m radius circle) downrange; the trajectory line is colored by altitude.

Experimental Drop Tests

Three instrumented drop-test campaigns validated the model, with the results summarized in Table 10:

Test	Geometry	v impact	Spin / Cone
Test 1	Asa1, 4 wings	10.33 m/s	306.8 rpm / 13.5°
Test 2	Asa2, 4 wings	10.05 m/s	306.9 rpm / 24.4°
Test 3	Asa3, 4 wings	12.90 m/s	372.0 rpm / 17.2°

Table 10 Instrumented drop-test results.

Test 2 (Asa2, 4 wings, wing area 122.28 cm^2) increased the wing area by 12% and reduced the impact velocity from 10.33 to 10.05 m/s, becoming the intermediate candidate. Test 3 (Asa3, 4 wings, 1000 m drop, simulated) reached steady autorotation at 12.90 m/s in 78.03 s, with $\theta_{eq} = 17.22°$ and 371.98 rpm, dissipating 99.15% of the impact energy.

Test 3 (Asa3, 4 wings) also validated the onboard electronics, with the SRAB instrumented to record rotation with the flight IMU; at a 1000 m drop it reached steady autorotation at 12.90 m/s in 78.03 s, with $\theta_{eq} = 17.22°$ and 371.98 rpm, dissipating 99.15% of the impact energy (ENMC consolidated data).

Geometry Comparison and Final Choice

A comparative simulation of Asa2 (4 wings) vs. Asa3 (2 wings) showed 10.05 vs. 13.33 m/s terminal velocity after the radius optimization. The Asa3 2-wing configuration (optimized aerodynamic radius 79.7 mm) was chosen as the final configuration for the mission: it reaches the 20/1.5 m/s design target with a 1.5 safety factor on the LASC 20 m/s limit, while offering lower mass and complexity than the 4-wing versions.

The choice between four and two wings is a direct trade-off between redundancy and simplicity. The 4-wing variants (Asa1, Asa2) provide geometric redundancy — a single lost wing degrades performance without preventing autorotation — and a lower terminal velocity (10.05 m/s), which increases the margin against the 20 m/s limit. However, they pay for this in mass, rotating inertia and mechanical complexity: each additional wing adds 3D-printed material, bonding area and a potential failure surface, and the higher rotating mass stores more kinetic energy at impact for the same descent velocity. The 2-wing Asa3 exploits the same autorotation physics with roughly half the wing area (58.7 cm^2 vs. 122.28 cm^2) and, being lighter, spins faster at equilibrium (477 rpm vs. 372 rpm for the 4-wing test geometry), which increases the gyroscopic stiffness that stabilizes the coning motion. At the cost of reduced geometric redundancy, it still sits 1.5× below the regulatory limit, and the drop-test evidence (Tests 1–3) confirms that deployment and steady autorotation are deterministic with the passive geometry — there are no hinges or actuators to jam, so the redundancy argument that motivates four wings is largely addressed by the mechanism's inherent robustness rather than by part count. The mission risk assessment (Appendix B) treats single-wing loss as a degraded-mode event bounded by the same 20 m/s limit, and the mechanical simplification of two wings directly supports the schedule-driven development goal of the project.

5. Conclusions and Lessons Learned

A. Conclusions

The Helike #213 mission shows that a bioinspired autorotative recovery system (SRAB) is a simpler, more reliable alternative to conventional parachutes for 1P PocketQubes. The SRAB is predicted to reach the LASC regulatory impact velocity with a fully passive mechanism, with no actuators, no pyrotechnics and no software dependency. The computational model, built on a 4th-order rigid body with BEM and a LEV correction, reproduces the measured drop-test behavior with enough fidelity for design decisions, and the full flight simulation, from the Dédalo launch vehicle to the SRAB descent under real GFS conditions, certifies the recovery. Against a conventional parachute, the SRAB shortens the descent by 36% and cuts the descent drift by 36%, which improves recovery accuracy at the cost of a higher, yet compliant, impact energy. Finally, the three-filter review (SCSM compliance, Murphy analysis, timeline discipline) proved to be an effective engineering framework: it caught the non-parachute notification requirement (REC 10.2.1), drove the redundancy decisions in firmware and hardware, and kept the project aligned with the competition deadlines.

Above all, the mission treats LASC not merely as a competition but as the operational proving ground of the SRAB technology: the September flight is the first full-scale, instrumented test of a passive samara-inspired recovery system in the conditions for which it was designed. The telemetry returned from the descent — rotation rate, descent velocity, impact conditions — is the evidence that will recalibrate the model parameters (e.g. C_{D0} and the LEV correction factor) and close the loop between simulation and flight, advancing the SRAB from a computational concept to a flight-proven technology available to future PocketQube missions.

B. Lessons Learned

LASC 2025 taught us that optimization without redundancy is a trap, and every subsystem of Helike carries a backup path: SD with LittleFS, LoRa with local storage, a hardware watchdog, and a passive recovery system that does not depend on electronics at all. We learned that the parachute is the weakest link in the recovery chain, and replacing it with a passive aerodynamic mechanism removes the most failure-prone element entirely. Iterative drop testing beat simulation alone: each test cycle, produced a measurable improvement in both the model and the design, and that simulation-validated iteration loop became the core of the project methodology. Compliance, we found, is a design input, not a formality; the SCSM requirements on kill switches, RBF, autonomy and notification shaped the hardware schematic and the firmware loop architecture from day one. On the team side, the three-filter reviews force decisions backed by evidence, which accelerates the transfer of knowledge from senior to new members: every filter review doubles as training for the next generation of the team.

6. References

A. Periodicals

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B. Reports, Theses, and Individual Papers

[5] Latin American Space Challenge, *Satellite Challenge Standards Manual (SCSM)*, 7th ed., Rev. 1, 2026.

[6] RocketPy Development Team, *RocketPy Documentation*, v1.2+, 2024–2026 [online database]. URL: https://docs.rocketpy.org/ [retrieved <data>].

Hazard Analysis Appendix

The following hazards were identified and mitigated by design and/or process, covering personnel safety during integration, launch and recovery operations. The complete hazard analysis is presented in Table 11:

Hazard ID & Title	Design Mitigation	Process Mitigation
Hazard 1 Battery thermal runaway	BMS with multiple protections covering overcharge, overdischarge, short-circuit, and overcurrent conditions.	Cells cycle-tested before integration. Battery charged only with approved charger, monitored during pre-launch, and RBF pin cuts all energy during transport.
Hazard 2 Stored energy during integration	Two independent mechanical kill switches (SW1/SW2) wired in parallel (NC) on main power line. RBF pin (SW3) in series cuts all energy when inserted.	Pre-integration verification procedure checks kill switch activation. RBF removed only after installation in the deployer.
Hazard 3 Sharp edges of SRAB wings	Final geometry features rounded leading edges.	Team wears handling gloves during integration. Recovery team briefed on wing handling.
Hazard 4 RF interference with other teams	Dedicated packet protocol with team ID, a reserved 915 MHz channel, and local storage for full data backup.	Coordinated frequency allocation during operations.
Hazard 5 Uncontrolled descent	Passive deployment with no actuators, two-wing configuration for predictable recovery, keeping nominal impact energy low (17.8 J).	Validated through three iterative simulation-and-test campaigns.

Table 11 Hazard analysis and mitigations.

Risk Assessment Appendix

Risk is quantified as $R = P \times S$, where P is the failure probability (1–3) and S the severity (1–3). Failure modes are organized by mission phase, following the ConOps defined in Section 2. The failure modes are ranked in the FMECA matrix (Table 12):

A. Risk Matrix (FMECA)

#	Failure Mode	Phase	P	S	R	Level
1	SRAB deployment failure (wings do not open)	Ejection	2	3	6	Critical
2	Battery failure (thermal runaway)	Flight	1	3	3	Medium
3	RBF pin does not cut all energy	Pre-launch	1	3	3	Medium
4	LoRa communication loss	Flight	2	2	4	High
5	SD storage failure	Flight	1	3	3	Medium
6	Structural break at launch	Launch	1	3	3	Medium
7	Kill switch not activated	Integration	1	3	3	Medium
8	RF interference (other teams)	Flight	2	2	4	High
9	GPS failure (no fix)	Post-landing	2	1	2	Low
10	SRAB works but satellite lands in inaccessible area	Recovery	1	2	2	Low

Table 12 Failure Mode, Effects, and Criticality Analysis (FMECA).

B. Mitigations

1. **SRAB deployment failure (Critical):** Passive design with no hinges; drop tests #1 and #2 validated deployment; 4 wings provide geometric redundancy — loss of one wing degrades performance but does not prevent recovery.
2. **Battery thermal runaway:** BMS with multiple protections; cell cycle testing.
3. **RBF pin:** RBF circuit in series with the main power rail.
4. **LoRa loss:** Local storage (SD) preserves all flight data; LoRa is support, not mission.
5. **SD failure:** Automatic LittleFS fallback on internal flash.
6. **Structural break:** ABS 3D-printed, walls $\geq$ 1.2 mm; load analysis.
7. **Kill switch:** 2 independent kill switches + pre-integration verification.
8. **RF interference:** Dedicated team ID protocol; 915 MHz channel reservation.
9. **GPS failure:** LoRa RSSI triangulation as backup; visual search.
10. **Inaccessible landing:** 3 GPS beacons + buzzer for localization.

C. Criticality Discussion

Failure mode #1 (SRAB deployment) is treated as critical and is the main driver of the extensive drop-test campaign. Drop tests #1 and #2 demonstrated that passive deployment works consistently without actuators. The remaining risk is bounded by the LASC regulatory limits and by the geometric redundancy of the wing arrangement. c

Engineering Drawings

This appendix contains the revision-controlled technical drawings and optional supporting material for the Helike #213 subsystems.

A. Drawing List

The revision-controlled drawing set is listed in Table 13. The electrical schematic of the subsystem is presented in Fig. 4:

Drawing / Document
Electrical schematic (KiCad, U1–U6, SW1–SW3)
SRAB wing geometry (Asa1/Asa2/Asa3)
PocketQube 1P mechanical structure
ESP32-C3 pinout
Integrated assembly

Table 13 Drawing list.

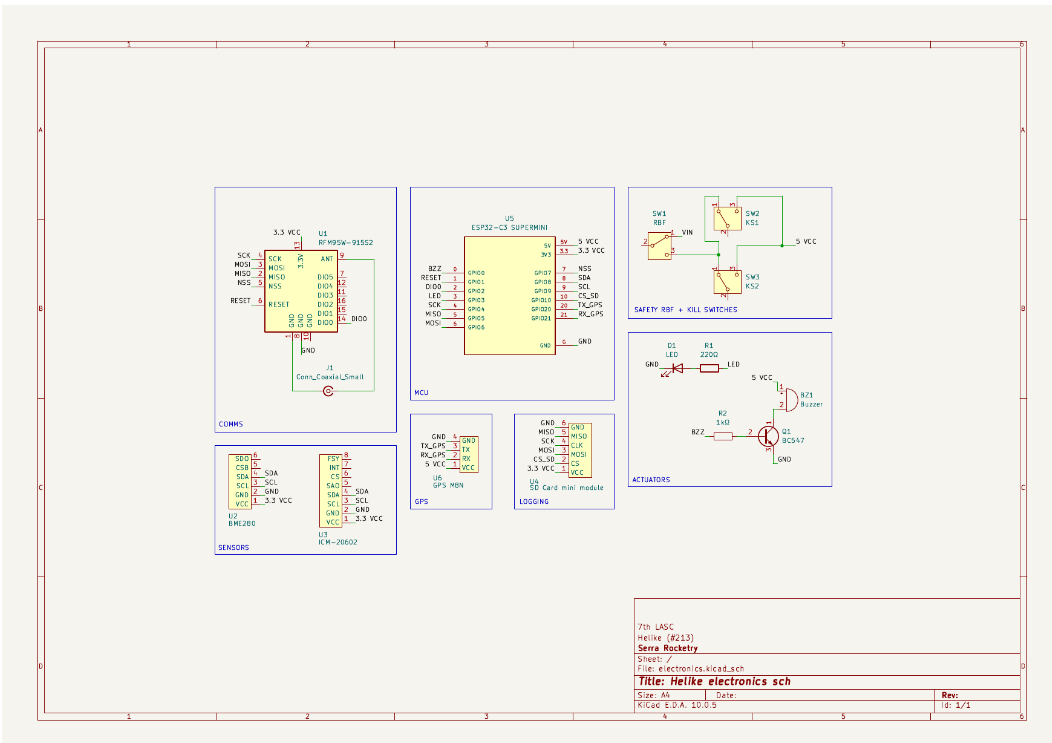

Fig. 4 Electrical schematic of Helike #213 (KiCad). U1: ESP32-C3 Super Mini; U2: LoRa RFM95W-915S2; U3: BME280; U4: ICM-20602; U5: GPS NEO-8M; U6: SD card module; SW1/SW2: kill switches (Z axis); SW3: RBF pin; J1: SMA antenna; D1/BZ1: status LED and buzzer.

B. Optional Appendices

SRAB Wing Geometry

The wing geometries evolved through three design iterations, driven by the simulation-to-field comparison loop described in Section 4:

- **Asa1** — 4 wings, first prototype: v_{impact} = 10.33 m/s, 444.9 rpm, 6.1°.
- **Asa2** — 4 wings, increased area and realignment: v_{impact} = 10.05 m/s, 439.1 rpm, 11.9° — candidate configuration.
- **Asa3** — 2 wings (mission final): reduced mass and complexity, $v \approx 20$ m/s, within the SCSM 20–45 m/s range.

Test Data

Computational Simulation

The 4th-order autorotation model, calibrated parameters, simulation scripts (RocketPy + SRAB extension) and the geometry heat-map are maintained in the repository under `extras/wing-analysis/`. The full-flight simulation (Dédalo + SRAB) is documented in the notebook `01_simulacao_dedalo.ipynb`.